TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2082 Kartik

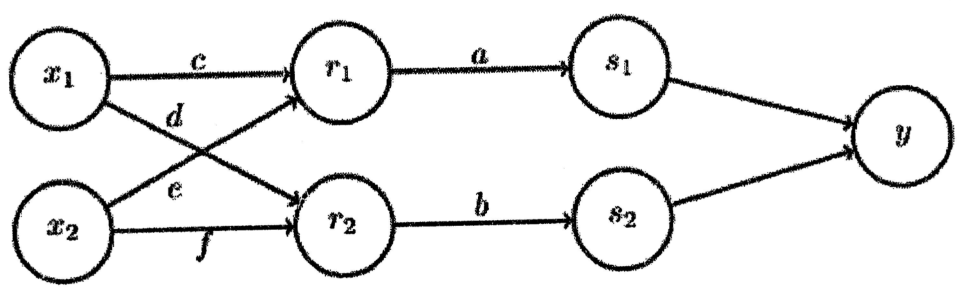
Exam.		Back	
Level	BE	Full Marks	80
Programme	BCT	Pass Marks	32
Year / Part	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (*CT 653*)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ *Attempt **All** questions.*
- ✓ *The figures in the margin indicate **Full Marks**.*
- ✓ *Assume suitable data if necessary.*

1. Define Artificial Intelligence. Design the PEAS information Automated Robot in a manufacturing plant. [2+3]
2. What are criteria for defining problem? Solve the following crypto-arithmetic algorithm: BASE + BALL = GAMES [2+6]
3. What do you mean by Searching and what are the types of Search methods? What are the drawbacks of Min-Max algorithm and explain how Alpha-Beta pruning algorithm solves it with a suitable example? [3+6]
4. a) Use resolution solve following problem: The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America has some missiles, and all of its missiles were sold to it by Colonel West, who is American. Prove That Col. West is criminal. [8]

   b) Define production system. Define and differentiate forward and backward chaining. [2+4]
5. Define term frame. Convert the following sentences into semantic network. Cat is a mammal. Cat has fur. Mammal has vertebra. Mammal is an animal. Bear is a mammal. Bear has fur. Whale is a mammal. Whale lives in water. Fish is an animal 10. Fish lives in water. [1+6]
6. What is fuzzy logic? Explain the steps involved in genetic algorithm with examples. [2+6]
7. Below is a neural network with weights a, b, c, d, e, and f. The input are x_1 and x_2. The first hidden layer computes $r_1 = \max(c.x_1+e.x_2, 0)$ and $r_2 = \max(d.x_1+f.x_2, 0)$. The second hidden layer and third layer computes s_1, s_2 and y using sigmoid activation function. Suppose the network has inputs $x_1 = -1$ and $x_2 = -1$. The weight values are a =1, b =1, c = 4, d = 1, e = 2 and f =2. Perform forward propagation. And update weight for second hidden layer for a and b. [8]



8. Define term expert system and expert system Shells. What are the characteristics that problem possesses are solved by expert system? Explain with example. [2+2+3]
9. Explain the McCulloch-Pitts model of a neuron with a diagram. Explain with example how to choose best activation function in particular project? [4+3]
10. What are the problem associated with NLP? List down the different steps involved in the natural language processing with suitable example. [2+5]

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